

Section 1

Edge Impulse Project Link: <https://studio.edgeimpulse.com/public/1003005/live>

1. The raw inputs are accX, accY, and accZ.
2. NN classifier features:
 - a. Type: Spectral features
 - b. Number of input features: 33
 - c. 5 feature names: accX RMS, accX Peak 1 Height, accY Peak 3 Height, accX Peak 1 Freq, accY RMS
3. Outputs 2 classes: "nominal" (for when the fan is operating normally) and "off" (for when the fan is off)
4. The anomaly detector uses spectral features as input, and there are a total of 4 used features:
 - a. accX RMS
 - b. accX Peak 1 Height
 - c. accY Spectral Power 2.0 - 5.0 Hz
 - d. accZ Spectral Power 2.0 - 5.0 Hz
- 5.

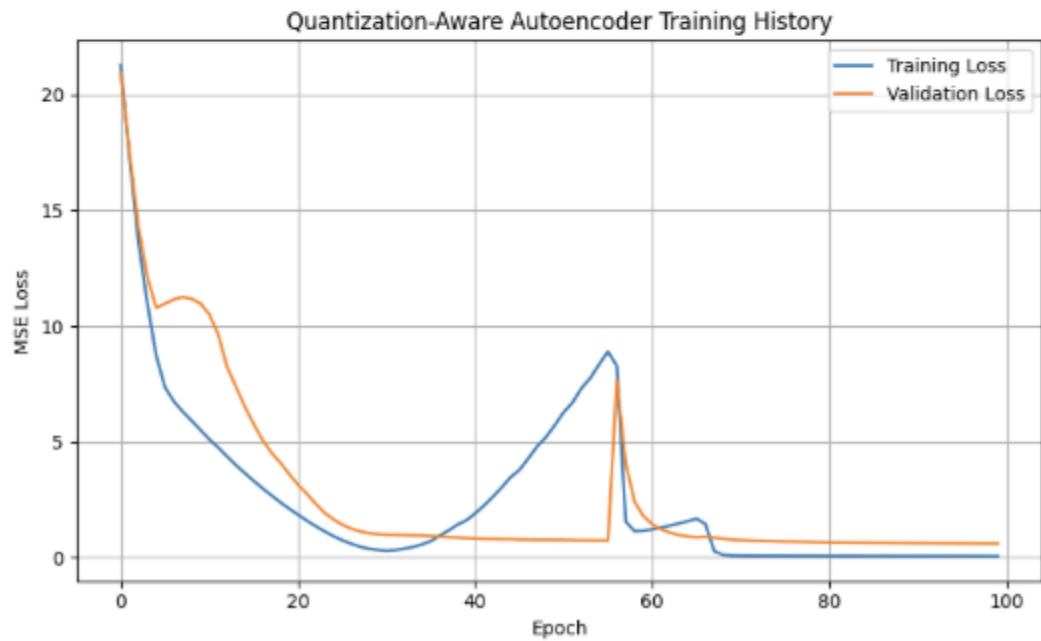
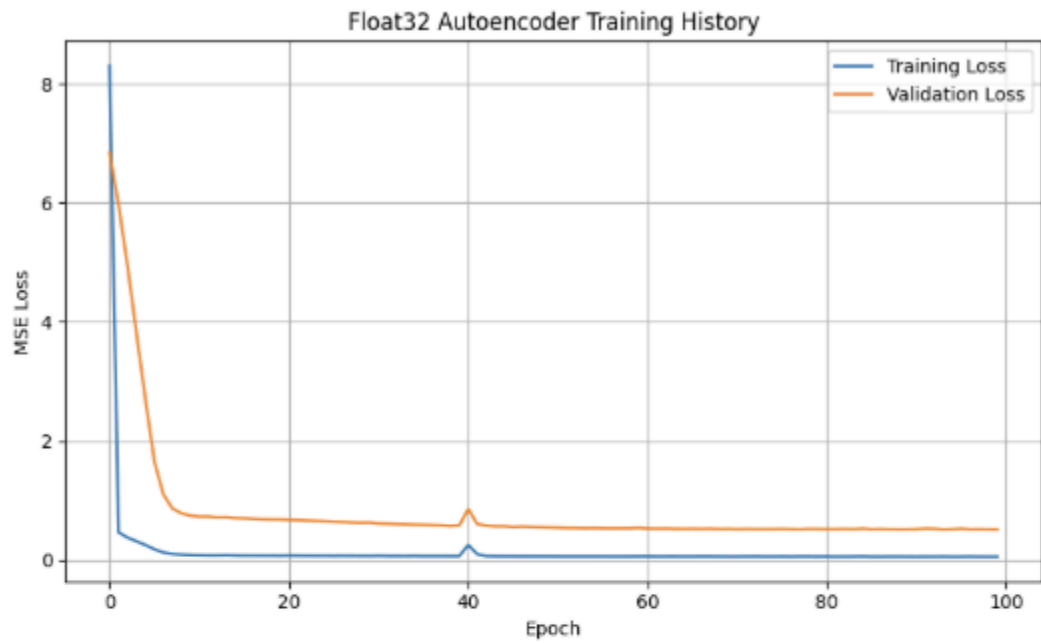
```
off: 0.000000
Anomaly prediction: 15.240551
Starting inferring in 2 seconds...
Sampling...
Timing: DDP 4s ms, inference 618 us, anomaly 588 us, postprocessing 50 us
#Classification predictions:
nominal: 0.437988
off: 0.462008
Anomaly prediction: 16.829048
Starting inferring in 2 seconds...
Sampling...
Timing: DDP 4s ms, inference 618 us, anomaly 587 us, postprocessing 49 us
#Classification predictions:
nominal: 0.437988
off: 0.562008
Anomaly prediction: 16.403193
Starting inferring in 2 seconds...
Sampling...
Timing: DDP 4s ms, inference 592 us, anomaly 586 us, postprocessing 105 us
#Classification predictions:
nominal: 0.437988
off: 0.562008
Anomaly prediction: 16.820322
Starting inferring in 2 seconds...
Sampling...
Timing: DDP 4s ms, inference 596 us, anomaly 524 us, postprocessing 968 us
#Classification predictions:
nominal: 0.462008
off: 0.537992
Anomaly prediction: 15.986072
Starting inferring in 2 seconds...
Sampling...
Timing: DDP 4s ms, inference 592 us, anomaly 524 us, postprocessing 58 us
#Classification predictions:
nominal: 0.462008
off: 0.537992
Anomaly prediction: 16.418923
Starting inferring in 2 seconds...
Sampling...
Timing: DDP 4s ms, inference 611 us, anomaly 525 us, postprocessing 49 us
#Classification predictions:
nominal: 0.437988
off: 0.562008
Anomaly prediction: 16.820322
Starting inferring in 2 seconds...
Sampling...
Timing: DDP 4s ms, inference 592 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
nominal: 0.462008
off: 0.537992
Anomaly prediction: 13.419226
Starting inferring in 2 seconds...
Sampling...
Timing: DDP 4s ms, inference 613 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
nominal: 0.718937
off: 0.281063
Anomaly prediction: 774.080926
Starting inferring in 2 seconds...
Sampling...
Timing: DDP 4s ms, inference 611 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
nominal: 0.462008
off: 0.537992
Anomaly prediction: 11.736388
Starting inferring in 2 seconds...
```

6. During deployment, the model predicted "off" with ~56% confidence and "nominal" with ~44% confidence, with an anomaly score of ~16. Since the board wasn't connected to a fan, the accelerometer data didn't closely match either training class, which explains the low confidence split and high anomaly score. The classifier output shouldn't always be trusted on its own; when the confidence values are this close, it might be because the model is uncertain. As for the anomaly score, a high score indicates that the input data falls outside the learned normal clusters, which may cause the classification result to be unreliable.

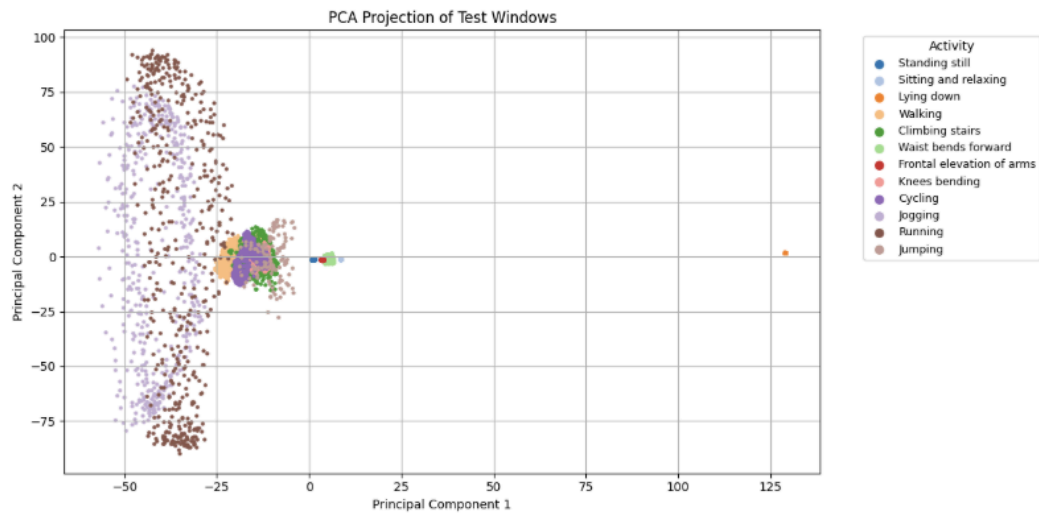
Section 2

Training and validation loss curves

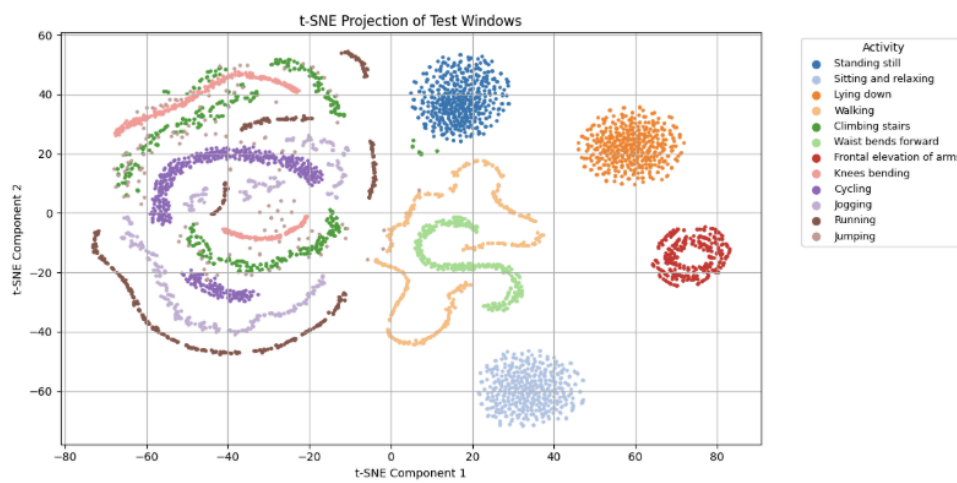
31/31 [=====] - 75 imgs/step - loss: 0.0043 - val_loss: 0.0119



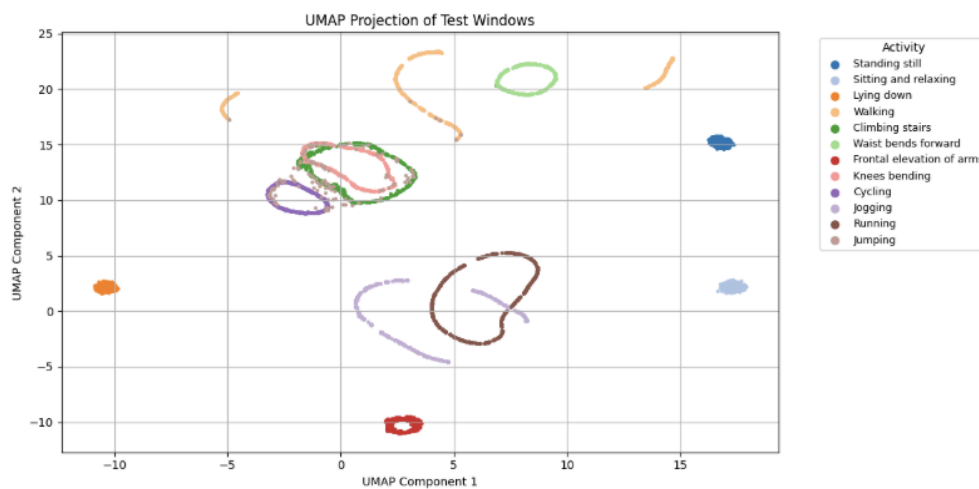
PCA visualization



t-SNE visualization



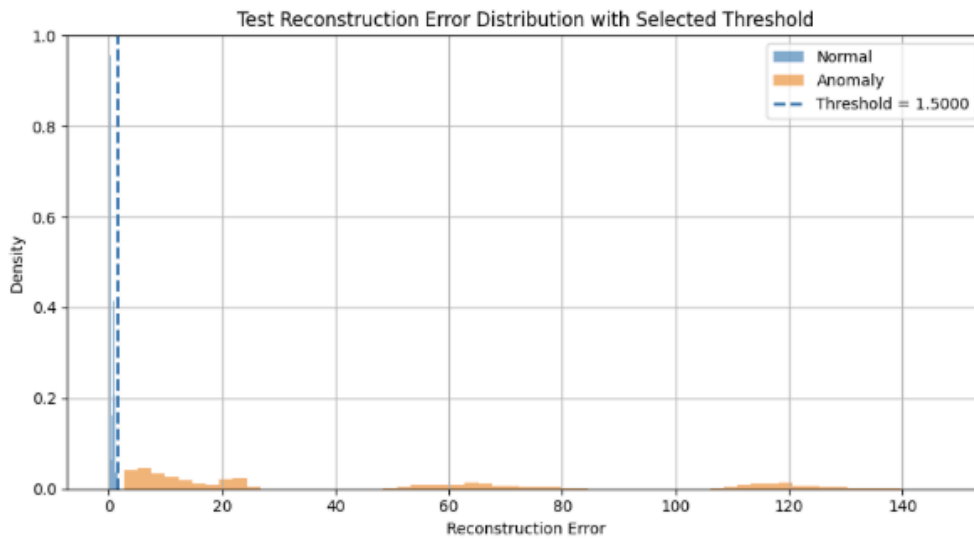
UMAP visualization



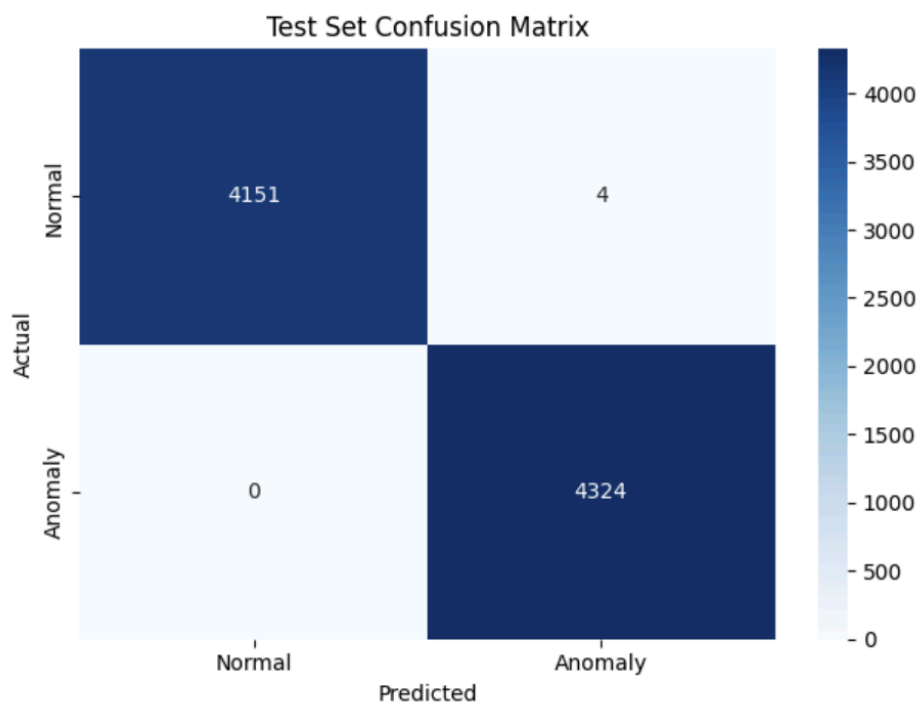
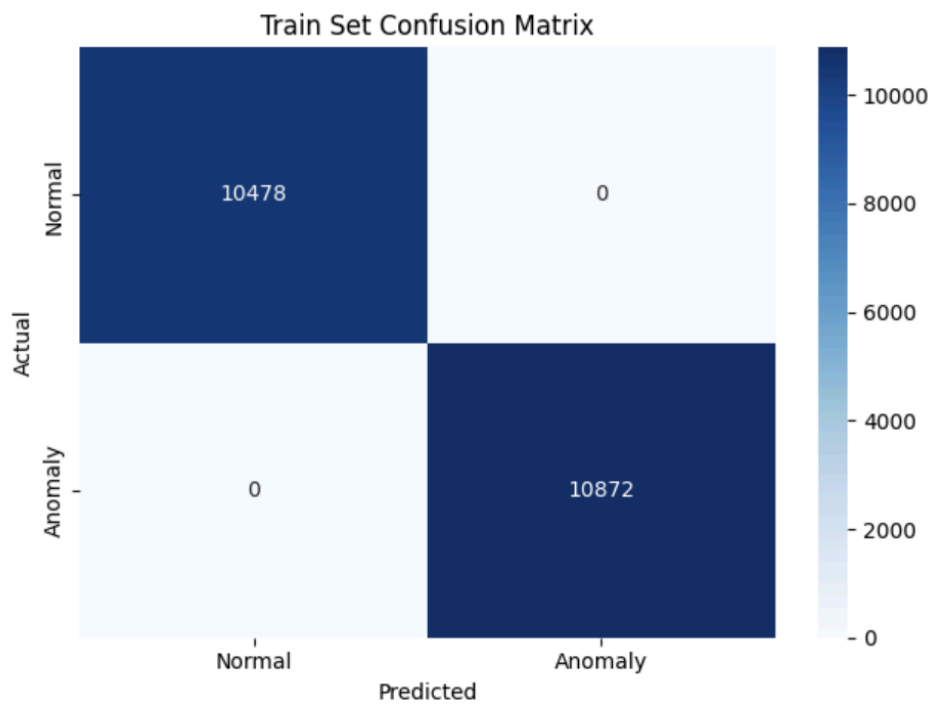
Reconstruction error distribution



Selected anomaly threshold



Confusion matrix



Classification report/evaluation metrics

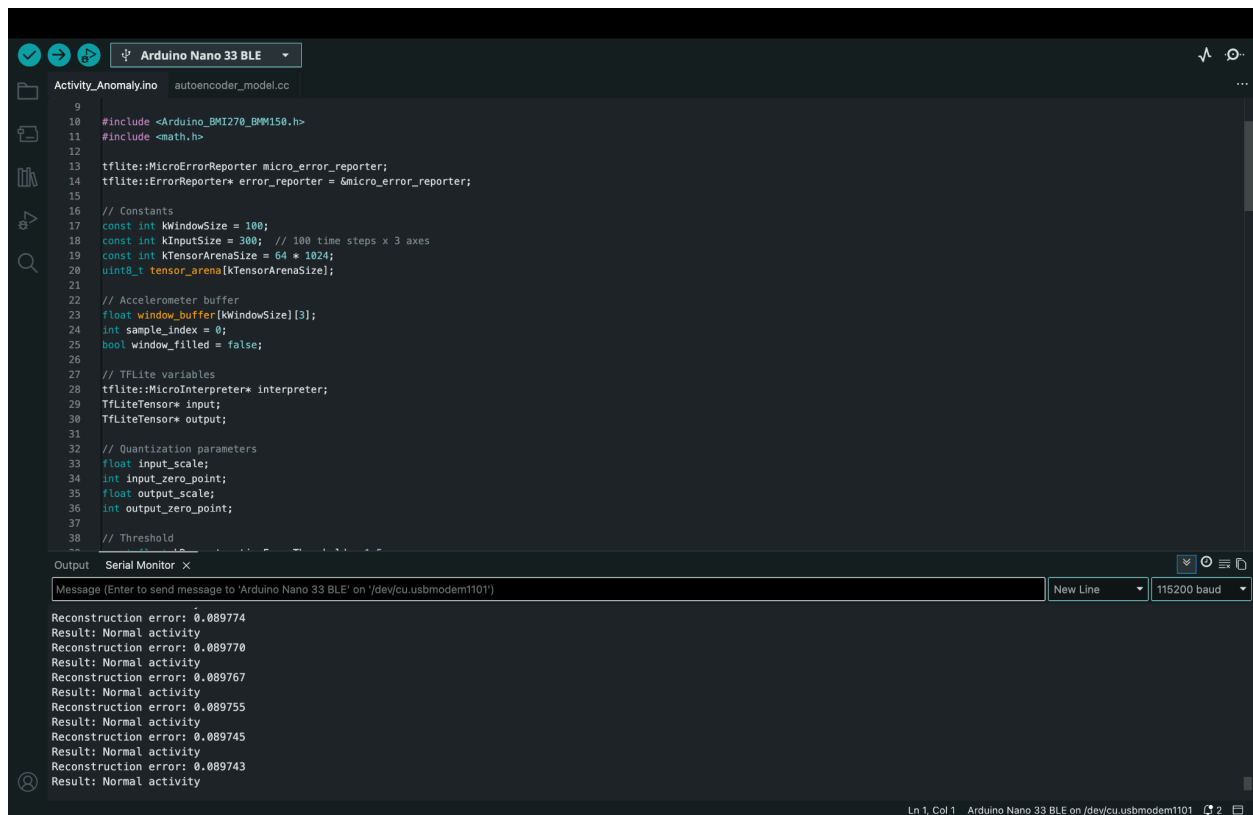
Train set evaluation

	precision	recall	f1-score	support
Normal (0)	1.00	1.00	1.00	10478
Anomaly (1)	1.00	1.00	1.00	10872
accuracy			1.00	21350
macro avg	1.00	1.00	1.00	21350
weighted avg	1.00	1.00	1.00	21350

Test set evaluation

	precision	recall	f1-score	support
Normal (0)	1.00	1.00	1.00	4155
Anomaly (1)	1.00	1.00	1.00	4324
accuracy			1.00	8479
macro avg	1.00	1.00	1.00	8479
weighted avg	1.00	1.00	1.00	8479

Serial monitor output



The screenshot shows an IDE window with the file `Activity_Anomaly.ino` open. The code is a C++ program for an Arduino Nano 33 BLE, using the `Arduino_BMI270_BMM150.h` library. It includes headers for `Arduino_BMI270_BMM150.h` and `math.h`. It defines a `tfLite::MicroErrorReporter` and a `tfLite::ErrorReporter*` pointer. It sets constants for `kWindowSize` (100), `kInputSize` (300), and `kTensorArenaSize` (64 * 1024). It declares a `uint8_t tensor_arena[kTensorArenaSize]` and a `float window_buffer[kWindowSize][3]`. It also declares `int sample_index` and a `bool window_filled` flag. It defines TFLite variables: `tfLite::MicroInterpreter* interpreter`, `TfLiteTensor* input`, and `TfLiteTensor* output`. It sets quantization parameters: `float input_scale`, `int input_zero_point`, `float output_scale`, and `int output_zero_point`. It also defines a `float Threshold`.

The serial monitor output shows the following messages:

```
Reconstruction error: 0.089774
Result: Normal activity
Reconstruction error: 0.089770
Result: Normal activity
Reconstruction error: 0.089767
Result: Normal activity
Reconstruction error: 0.089755
Result: Normal activity
Reconstruction error: 0.089745
Result: Normal activity
Reconstruction error: 0.089743
Result: Normal activity
```